Food selection based on high total antioxidant capacity improves endothelial function in a low cardiovascular risk population.

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BACKGROUND AND AIMS: Oxidative stress has been advocated as a major cause for cardiovascular disease (CVD), and low plasma antioxidant concentrations are associated with endothelial dysfunction, the first step towards atherosclerosis. However, although the antioxidant content in fruits and vegetables may explain at least in part their protective effect against CVD, supplementation with antioxidant vitamins fails to improve endothelial function and reduce CVD risk. The aim of this study was to investigate the impact of a diet rich in antioxidants on endothelial function measured by flow-mediated dilatation (FMD) in volunteers at low cardiovascular risk. METHODS AND RESULTS: In a crossover trial, 24 subjects (13 women, mean age 61 ± 3 years), received, in a randomised order, a 14-day high (HT) and a 14-day low (LT) antioxidant diets, with a 2-week wash-out (WO) in between. Both diets were comparable in daily portions of fruits and vegetables, and in alcohol, fibre and macronutrient intake, but differed in their total antioxidant capacity. Before and after each diet, anthropometrics, blood pressure, fasting plasma glucose, lipid profile, hepatic enzymes, circulating antioxidant concentrations, high sensitivity C-reactive protein (hs-CRP) and FMD were assessed. FMD increased significantly during the HT diet compared to the LT ($p < 0.000$). FMD values were 2.3% higher after HT compared with LT ($p < 0.001$) after adjustment for age, gender and diet order. α-tocopherol increased significantly ($p < 0.05$) and hs-CRP and of γ-glutamyltranspeptidase decreased significantly ($p < 0.05$ and $p < 0.01$, respectively) during the HT diet, compared with the LT diet. CONCLUSIONS: A short-term HT diet improves endothelial function in volunteers at low cardiovascular risk, which may further reduce their risk of CVD.